

Project Management – Module 4 Notes

Overview

These notes cover the fourth and final module of Project Management: reducing project duration through crashing, project risk management, project evaluation, project progress and performance management, project closure and oversight, and an introduction to project management software. The notes are written for exam preparation with clear definitions, formulas, comparison tables, and likely exam questions.

1. Reducing Project Duration

Reducing project duration means shortening the total time required to complete a project. This becomes necessary due to client demands, contractual penalties, competitive pressure, strategic urgency, or schedule overruns.[web:51][web:50]

Reasons for reducing project duration

- Market or competitive pressure to launch earlier.[web:51]
- Contractual deadlines with penalty clauses.[web:52]
- Reallocation of resources to another project.[web:51]
- Schedule recovery after delays.[web:50]
- External factors such as regulatory or seasonal deadlines.[web:51]

Options for reducing project duration

- Adding more resources to critical activities.[web:52]
- Outsourcing project activities to specialized parties.[web:52]
- Working overtime or extending working shifts.[web:52]
- Reducing project scope after consulting stakeholders.[web:52]
- Phasing the project deliverables so parts are released earlier.[web:52]
- Fast-tracking: overlapping phases or activities that were originally sequential.[web:63]

2. Crashing Project Activities

Project crashing is the process of shortening the project schedule duration for the least additional cost by adding extra resources to critical path activities.[web:50][web:53]

Key concepts

- **Normal time:** The standard duration estimated under normal conditions.[web:52]
- **Crash time:** The minimum possible time to complete an activity after applying extra resources.[web:52]
- **Normal cost:** The cost of completing an activity in its normal time.[web:52]
- **Crash cost:** The higher cost incurred when completing the activity in crash time.[web:52]
- **Crash cost per unit time:** The additional cost for reducing one unit of duration.[web:53]

Crash cost per unit time formula

$$\text{Crash cost per unit} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash time}}$$

The activity with the **smallest crash cost per unit of time** on the critical path should be crashed first.[web:53]

Steps in crashing a project

1. Start with all activities at normal time and normal cost.[web:52]
2. Find the critical path and total project duration.[web:52]
3. Calculate crash cost per unit of time for each critical activity.[web:53]
4. Select the critical activity with the lowest crash cost per unit.[web:53]
5. Crash that activity by one time unit and recalculate total cost and duration.[web:53]
6. Update the critical path (it may change after crashing).[web:53]
7. Repeat until target duration is reached or further crashing is not worthwhile.[web:50]
[web:53]

Important rules for crashing

- Only activities on the critical path can reduce overall project duration.[web:52][web:53]
- If there are multiple critical paths, one activity from **each** must be crashed simultaneously.
[web:53]
- Crashing beyond the crash time is not possible.[web:52]
- Total project cost = direct cost + indirect cost; reduce total cost by finding the optimal crash point.[web:52]

Crashing vs Fast-Tracking

Basis	Crashing	Fast-Tracking
Method	Add more resources [web:50][web:63]	Overlap sequential activities [web:63]
Cost	Increases cost [web:52][web:53]	May increase risk but not necessarily cost [web:63]

Basis	Crashing	Fast-Tracking
Risk	Lower risk if resources are available [web:63]	Higher risk due to parallel work [web:63]
Scope	Scope unchanged [web:52][web:63]	Scope unchanged [web:63]

3. Project Risk Management

Project risk management is the systematic process of identifying, analyzing, planning responses to, and controlling risks in order to protect project objectives from uncertainty. [web:54][web:59]

What is risk?

Risk is an uncertain event or condition that, if it occurs, has a positive or negative effect on project objectives including scope, schedule, cost, and quality. Risk can be a **threat** (negative) or an **opportunity** (positive).[web:54]

Steps in project risk management

Step 1: Risk Identification

- Identify all potential risks that might affect the project.[web:54]
- Tools: brainstorming, checklists, expert interviews, SWOT analysis, historical records. [web:54][web:59]
- Output: **Risk Register** — a document listing all identified risks.[web:54]

Step 2: Risk Analysis

Qualitative risk analysis: Prioritize risks using probability and impact rating. Each risk is assessed for its likelihood and consequence on a scale (e.g., Low, Medium, High).[web:54] [web:59]

Quantitative risk analysis: Use numerical tools to estimate the probability and financial impact of risks. Tools include Monte Carlo simulation, decision tree analysis, and sensitivity analysis. [web:54]

Step 3: Risk Response Planning

Develop strategies to reduce threats and enhance opportunities:[web:54][web:59]

Strategy	For Threats	Meaning
Avoid [web:54]	Yes	Change the plan to eliminate the risk entirely
Transfer [web:54]	Yes	Shift the financial impact to a third party (insurance, contracts)
Mitigate [web:54]	Yes	Reduce probability or impact

Strategy	For Threats	Meaning
Accept [web:54]	Yes	Acknowledge the risk and prepare a contingency
Exploit [web:54]	Opportunities	Take advantage of positive risk

Step 4: Risk Monitoring and Control

- Track identified risks and identify new ones throughout the project.[web:54]
- Conduct risk audits and review risk register periodically.[web:54]
- Implement contingency plans when risks occur.[web:54]

4. Project Evaluation

Project evaluation is the systematic assessment of project objectives, processes, performance, and outcomes. It helps determine whether a project is achieving its intended goals and provides a basis for decision-making and improvement.[web:55][web:60]

Types of evaluation

- **Pre-project evaluation:** Assesses viability before the project starts.
- **Mid-project evaluation:** Checks whether the project is on track during execution.
- **Post-project evaluation:** Reviews outcomes after closure to assess impact and lessons.

Criteria for evaluation

- Achievement of scope and deliverables.
- Adherence to schedule and cost baselines.
- Quality of outputs and stakeholder satisfaction.[web:60]
- Efficiency of resource use.
- Risk management effectiveness.[web:60]

5. Project Progress and Performance Management

Project progress monitoring

Progress monitoring tracks the actual status of work against the planned baseline. It answers questions like: Is the project on schedule? Is spending on track? Are deliverables being produced correctly?[web:60]

Earned Value Management (EVM)

EVM is the most widely used technique for measuring project performance. It integrates scope, time, and cost to provide an objective view of project health.[web:55][web:60]

Key EVM terms and formulas

Term	Abbreviation	Meaning	Formula
Planned Value	PV	Budgeted cost of work scheduled	Given in baseline plan [web:60]
Earned Value	EV	Budgeted cost of work actually performed	% Complete × Budget [web:55]
Actual Cost	AC	Actual cost incurred to date	From accounting records [web:60]
Cost Variance	CV	Difference between EV and AC	$CV = EV - AC$ [web:60]
Schedule Variance	SV	Difference between EV and PV	$SV = EV - PV$ [web:60]
Cost Performance Index	CPI	Efficiency of cost use	$CPI = EV / AC$ [web:60]
Schedule Performance Index	SPI	Efficiency of schedule use	$SPI = EV / PV$ [web:60]

Interpretation

- $CV > 0$: Under budget; $CV < 0$: Over budget.[web:60]
- $SV > 0$: Ahead of schedule; $SV < 0$: Behind schedule.[web:60]
- $CPI > 1$: Good cost efficiency; $CPI < 1$: Poor cost efficiency.[web:60]
- $SPI > 1$: Ahead of schedule; $SPI < 1$: Behind schedule.[web:60]

Performance reporting

- Regular performance reports compare actual work, cost, and time against the plan.
- Reports help stakeholders stay informed and enable timely corrective action.[web:60]

6. Project Closure and Oversight

Project closure is the formal final phase in which all project activities are completed, deliverables are handed over, contracts are closed, resources are released, and project records are finalized and archived.[web:56][web:61]

Why project closure is important

- Ensures all deliverables are formally accepted by the client.[web:56]
- Releases resources for other projects.[web:61]
- Documents lessons learned for future projects.[web:56][web:61]

- Reduces organizational risk from unresolved contracts or commitments.[web:56]
- Provides a complete and auditable record of the project.[web:61]

Steps in project closure

1. **Verify scope:** Confirm all deliverables have been completed and accepted.[web:61]
2. **Transfer deliverables:** Hand over project outputs to the client or operations team.[web:61]
3. **Close contracts:** Formally close all supplier and subcontractor agreements.[web:61]
4. **Release resources:** Reassign project team members, equipment, and materials.[web:61]
5. **Document lessons learned:** Record what went well, what went wrong, and how to improve.
[web:56][web:61]
6. **Prepare close-out report:** Summarize project performance including cost, schedule, quality, and risk outcomes.[web:61]
7. **Archive project records:** Store documentation for future reference and audit.[web:56]
[web:61]
8. **Celebrate and formally close:** Hold a close-out meeting and officially announce project completion.[web:61]

Close-out report contents

- Summary of project objectives and deliverables.[web:61]
- Performance data on schedule, cost, and quality.[web:61]
- Description of risks encountered and how they were handled.[web:61]
- Lessons learned documentation.[web:61]
- Final recommendations for related future projects.[web:61]

Project oversight

Project oversight means the ongoing review and governance of a project to ensure it remains aligned with organizational goals, complies with policies, and delivers its intended value.

Oversight mechanisms include:

- Steering committee reviews.
- Independent project audits.
- Phase gate reviews before proceeding to the next stage.
- Portfolio-level monitoring of multiple projects.

7. Introduction to Project Management Software

Project management software provides digital tools that support planning, scheduling, resource management, communication, collaboration, cost tracking, and reporting for projects.[web:57][web:62]

Why use project management software?

- Automates scheduling and network diagram creation.[web:57]
- Centralizes information and improves team collaboration.[web:62]
- Tracks costs, time, and performance in real time.[web:57]
- Generates Gantt charts, PERT/CPM charts, and progress reports automatically.[web:62]
- Manages resource allocation and workload balancing.[web:57]

Categories of project management software

Category	Description	Example tools
Desktop software	Installed on a local system, powerful for scheduling	MS Project [web:62]
Cloud-based tools	Accessible online, supports team collaboration	Asana, Trello, monday.com [web:57][web:62]
Enterprise tools	Full-featured, integrated with ERP/finance systems	Oracle Primavera, Jira [web:62]
Open-source tools	Free and customizable	GanttProject, OpenProject [web:62]

Popular project management tools

Microsoft Project

- Industry-standard software for scheduling and resource management.
- Supports Gantt charts, critical path analysis, and EVM.
- Widely used in construction, IT, and infrastructure projects.[web:62]

Primavera P6 (Oracle)

- Enterprise-grade software designed for large and complex projects.
- Strongly used in engineering, construction, and government projects.
- Supports multi-project planning and portfolio management.[web:62]

Asana

- Cloud-based, user-friendly collaboration and task management tool.
- Suited for team coordination, task assignment, and progress tracking.[web:57][web:62]

Trello

- Visual kanban board tool for tracking task status and workflow.
- Simple and easy to use; suited for small projects and agile teams.[web:62]

Jira

- Popular for software development and IT projects.
- Supports agile, scrum, and kanban methodologies.[web:62]

monday.com

- Flexible cloud platform for project tracking, dashboards, and reporting.
- Suitable for teams of various sizes.[web:57]

Key features to look for in PM software

- Task and activity scheduling.
- Gantt chart and network diagram creation.
- Resource allocation and workload management.
- Budget and cost tracking.
- Real-time progress reporting and dashboards.
- Team communication and document sharing.
- Risk log and issue tracking.

Quick Revision Summary

- Crashing reduces project duration by adding resources to critical activities at extra cost.
[web:52][web:53]
- Always crash the critical activity with the **lowest crash cost per unit of time** first.[web:53]
- Fast-tracking overlaps activities; crashing adds resources.[web:63]
- Risk management steps: Identify → Analyze → Plan response → Monitor and control.[web:54]
[web:59]
- EVM integrates scope, cost, and time to measure project performance.[web:60]
- Key EVM formulas: $CV = EV - AC$, $SV = EV - PV$, $CPI = EV/AC$, $SPI = EV/PV$.
[web:60]
- Project closure includes scope verification, deliverable transfer, contract closure, lessons learned, and archiving.[web:56][web:61]
- PM software ranges from desktop tools like MS Project to cloud platforms like Asana, Trello, and Jira.[web:57][web:62]

Likely Exam Questions

1. What is project crashing? Explain the steps involved in crashing a project schedule.
2. Distinguish between crashing and fast-tracking.
3. Explain the process of project risk management.
4. Discuss Earned Value Management (EVM) with relevant formulas.
5. What is project evaluation? Explain its types and criteria.
6. Describe the process and importance of project closure.
7. Write a note on any two popular project management software tools.
8. Explain the time-cost trade-off with reference to crashing.

References

1. PMI, "Performing crash analysis: alternative cost-schedule scenarios." [web:49]
2. [ProjectManagers.org](#), "Crashing the Project Schedule: A Step-by-Step Guide." [web:50]
3. BITS Pilani, "Chapter 9: Reducing Project Duration (PDF)." [web:51]
4. MCRHRDI, "Crashing – Project Management Lecture Notes (PDF)." [web:52]
5. YouTube / Operations University, "Project Crashing Explained – Critical Path Method (CPM)." [web:53]
6. [IndustrialAudit.com](#), "Risk Management and Risk Mitigation." [web:54]
7. DataGuard, "Understanding Risk Assessment Methodology." [web:59]
8. [ProjectManagement.com](#), "What is the Earned Value (EV) of a Project?" [web:55]
9. PMI, "EVM and CPM: Evaluate Project Performance (Earned Schedule)." [web:60]
10. PMI, "Importance of the Closing Process Group." [web:56]
11. eCampus Ontario, "11.5 Lessons Learned – Essentials of Project Management." [web:61]
12. [project-management.com](#), "Top 10 Project Management Software for 2026." [web:57]
13. PCMag, "The Best Project Management Software for 2026." [web:62]
14. Project Management Academy, "PMP Exam Prep: Fast-Tracking vs Crash Schedule." [web:63]